

Lab 2 Report

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DFCS 605: Digital Forensics and Cyber Investigation Foundations

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Project: Digital Forensic Investigation

1. Summary

For this lab, I created a rich text file document in WordPad. Once this was done, a copy was made by right-clicking the document on the Desktop and pasting to an empty space on the Desktop. The copy was renamed and never opened in WordPad. Both documents were then analysed in FTK Imager where it was noted that the copy of the original had some timestamp discrepancies. However, both the MD5 and SHA1 hashes of both matched. After this was done, an analysis was done in RegEdit where it was noted that only the original existed in the WordPad registry edit view.

2. List of Evidence

The evidence analysed in this case included the items listed in Table 1.

Table 1: Evidence

Item / File	Hashes
Simpson.rtf	MD5: fc5dc6d9a3dcf383557327e164b05bda SHA1: a11c1b4028190c8f5b9220b3c044283f934a8c9e
Robert.rtf	MD5: fc5dc6d9a3dcf383557327e164b05bda SHA1: a11c1b4028190c8f5b9220b3c044283f934a8c9e

3. List of Forensic Tools

The tools listed in Table 2 were used to analyze evidence.

Table 2: Forensic Tools

Tool	Version Number
FTK Imager	4.7.2.11

4. Explanation of Work

First, WordPad was used to create a sample document named *Simpson.rtf*. This was saved to the Desktop. The document was copied and then pasted on to the Desktop. This copy was renamed *Robert.rtf*. It is important to note that *Robert.rtf* was not opened in WordPad or any other software prior to being viewed in FTK Imager. Once these two documents were made, they both were viewed in FTK Imager. For *Simpson.rtf*, the “Date Accessed”, “Date Created”, and “Date Modified” were all “2/2/2026 9:33:40 PM”. Note that date and time are from the virtual machine. For *Robert.rtf*, the “Date Accessed” and “Date Created” were “2/2/2026 9:34:24 PM” but the “Date Modified” was “2/2/2026 9:33:40”. The discrepancy is because *Robert.rtf* is a copy of the original document, *Simpson.rtf*. That is why it has an earlier modified time, as it was modified from the original (Briar, 2016).

Simpson.rtf Metadata

Date Accessed	2/2/2026 9:33:40 PM
Date Created	2/2/2026 9:33:40 PM
Date Modified	2/2/2026 9:33:40 PM

Robert.rtf Metadata

Next, RegEdit was opened and I went to HKEY_CURRENT_USER > Software > Microsoft > Windows > CurrentVersion > Applets > Wordpad > Recent File List. It is important to note that only “Simpson.rtf” appears here because “Robert.rtf” was never opened in WordPad.

Figure 1 shows the file signature for the file named *alpha.jpg* as seen in FTK Imager.

The screenshot displays the AccessData FTK Imager 4.7.2.11 interface. The main window shows a file tree on the left and a file list on the right. The Properties pane is open on the left, showing details for the selected file, 'Simpson.utf'. The File List pane on the right shows a list of files and folders within the image, including 'Alpha.jpg', 'desktop.ini', 'Evidence000hashes.csv', 'Forensicimage.adf', 'Forensicimage.adf.bak', 'ghidra.link', 'hashes.csv', 'Programs', 'Robert.utf', 'Simpson.utf', 'Simpson.txt', and 'WSL Ubuntu 1804.iso'. The Properties pane shows details for 'Simpson.utf', including its file class, size, physical size, date accessed, date created, date modified, encrypted status, compressed status, actual file, DOS attributes, and NTFS information.

Figure 1: Properties pane (left) of Simpson.utf

AccessData FTK Imager 4.7.2.11

File View Mode Help

Evidence Tree

- Administrative
 - cache
 - desktop.ini
 - ghidra
 - hashes.csv
 - Programs
 - Robert.utf
 - Simpson.utf
 - Simpson.txt
 - WSL Ubuntu 1804.iso

Properties

Simpson.utf

Name: Simpson.utf
File Class: Regular File
File Size: 288
Physical Size: 272
Date Accessed: 3/22/2024 9:33:40 PM
Date Created: 3/22/2024 9:33:40 PM
Date Modified: 3/22/2024 9:33:40 PM
Encrypted: False
Compressed: False
Actual File: True

DOS Attributes

Hidden: False
System: False
Read only: False
Archive: True

NTFS Information

MFT Record Number: 540,945 (03307600)
Data Changed (MFT): 3/22/2024 9:33:40 PM
Resident: True
Offline: False
Sparse: False
Temporary: False
Owner SID: S-1-5-21-37-71287537-220290000-1167480185-3011

Properties: Hex Value Inspector | Custom Content Sources

For User Guide, press F1

File List

Name	Size	Type	Date Modified
Alpha.jpg	32,288 (32 KB)	Regular File	1/14/2024 9:14:05 PM
desktop.ini	288 (1 KB)	Regular File	12/13/2023 9:09:34 PM
Evidence000hashes.csv	148 (1 KB)	Regular File	9/25/2023 4:18:17 PM
Forensicimage.adf	3,173,547 (3.14 MB)	Regular File	1/14/2024 9:13:34 PM
Forensicimage.adf.bak	363 (1 KB)	Regular File	1/09/2024 7:40:05 PM
ghidra.link	2,587 (3 KB)	Regular File	3/10/2024 12:59:40 AM
hashes.csv	1,859 (2 KB)	Regular File	1/13/2024 1:40:31 PM
Programs	2,339 (3 KB)	Regular File	9/16/2023 6:02:31 PM
Robert.utf	248 (1 KB)	Regular File	2/2/2024 9:33:40 PM
Simpson.utf	288 (1 KB)	Regular File	2/2/2024 9:33:40 PM
Simpson.txt	36 (1 KB)	Regular File	1/14/2024 9:14:04 PM
WSL Ubuntu 1804.iso	1,197 (1 KB)	Regular File	9/27/2023 14:59:59 PM

```
(\rft1\anal\enricopgi2024\def0\analysis\comp\defLang1033\fonttbl{\f0\fmil\Schaefer0 Calibri;})
(\*generator Microsoft Word 10.0.277053)viewwdd\vol\
\pard\wa2009\al276\alimult\tl\fo\fa22\lang9 Simpson\par
2/2/2024 4:52\par
Ref\par
I enjoy reading in my free time. \par
}
```

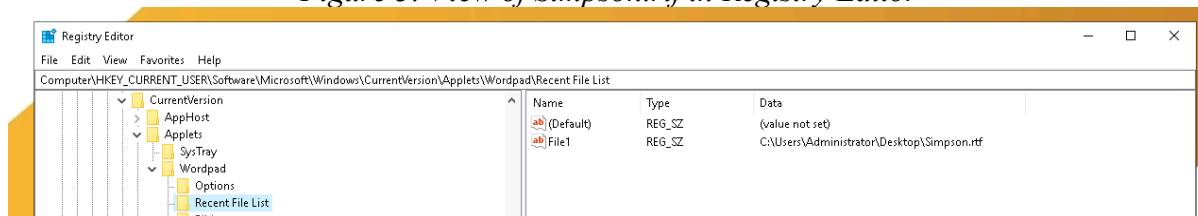
The screenshot displays the AccessData FTK Imager 4.7.2.11 interface. The main window is divided into several panes:

- Evidence Tree:** Shows the directory structure of the image. The 'Administrator' folder is expanded, showing subfolders like 'cache', 'desktop.ini', 'ghidra', 'icsoft', 'idlcrc', 'junique', 'splunk', and 'ssh'.
- Properties:** Displays the properties of the selected file, 'Robert.rtf'. The properties include Name, File Class, File Size, Physical Size, Date Accessed, Date Created, Date Modified, Encrypted, Compressed, Actual File, and DOS Attributes.
- File List:** A table listing files and their properties. The selected file is 'Robert.rtf'.
- Hex View:** Displays the raw data of the selected file. The text is encoded in UTF-16.

Name	Size	Type	Date Modified
lock.Evidence001hashes.csv#	122 (1 KB)	Regular File	1/14/2026 2:14:05 PM
alpha.jpg	32,290 (32 KB)	Regular File	12/13/2023 9:05:34 PM
desktop.ini	282 (1 KB)	Regular File	9/25/2023 4:18:17 PM
Evidence001hashes.csv	169 (1 KB)	Regular File	1/14/2026 2:13:34 PM
ForensicImage.ad1	3,217,547 (3.14...)	Regular File	1/28/2026 7:48:05 PM
ForensicImage.ad1.btt	963 (1 KB)	Regular File	1/28/2026 7:48:05 PM
ghidra.link	2,587 (3 KB)	Regular File	3/10/2022 12:59:43 AM
hashes.csv	1,959 (2 KB)	Regular File	1/13/2026 1:40:32 PM
Proxy.link	2,399 (3 KB)	Regular File	9/8/2020 6:00:23 PM
Robert.rtf	268 (1 KB)	Regular File	2/2/2026 9:33:40 PM
Simpson.rtf	260 (1 KB)	Regular File	2/2/2026 9:33:40 PM
Simpson.txt	36 (1 KB)	Regular File	1/14/2026 1:14:04 PM
WSL Ubuntu 1804.link	1,197 (2 KB)	Regular File	9/27/2023 1:47:59 PM

```
{\rtf1\ansi\ansicpg1252\deff0\nouicompat\deflang1033{\fonttbl{\f0\fmnil\chse(\*\generator Riched20 10.0.17763)\viewkind4\uc1\nord\sa200\sl276\slmult1\fo\fs22\lang9 Simpson\par2/2/2026 4:32\parRed\parI enjoy reading in my free time. \par}
```

Figure 3: View of Simpson.rtf in Registry Editor



5.1. Metadata for Files

Simpson.rtf Metadata

Date Accessed	2/2/2026 9:33:40 PM
Date Created	2/2/2026 9:33:40 PM
Date Modified	2/2/2026 9:33:40 PM

Robert.rtf Metadata

Date Accessed	2/2/2026 9:34:24 PM
Date Created	2/2/2026 9:34:24 PM
Date Modified	2/2/2026 9:33:40 PM

5.2 Registry Data

(Default)	REG_SZ	Value not set
File 1	REG_SZ	C:\Users\Administrator\Desktop\Simpson.rtf

6. Conclusions

To conclude, both the MD5 and SHA1 hashes matched between the original rich text file document and the copy. However, the copy had the same Date Modified timestamp as the original, but the Date Accessed and Created were unique to itself. Lastly, when looking at the registry editor for WordPad, only the original was present because the copy was never opened in WordPad.

7. References

Briar, J, & Shavers, B. *Hiding behind the keyboard: uncovering covert communication*

methods with forensic analysis. Amsterdam, Boston: Elsevier, 2016.